A reconductoring approach for efficient long-term planning of the Ecuadorian power grid

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***Abstract*—Since 2023, Ecuador has faced significant challenges in the energy sector due to regional droughts impacting hydroelectric capacity and increasing reliance on energy imports. In this context, planning the expansion of the infrastructure that generates and transmits electricity to end consumers should focus on strategies that optimize the use of available resources. This paper presents a Mixed Integer Linear Programming (MILP) approach for multistage combined electricity generation and transmission planning (GTEP). The model employs a reconductoring strategy for existing 230 kV transmission lines, integrating higher-capacity conductors. The practicality and effectiveness of the model are validated through a case study of Ecuador's power system, which features a 21-bus system. The results confirm that the model provides cost-efficient solutions, with a projected investment of $4417.40 MUSD to diversify energy sources and reduce reliance on imports, a crucial step towards achieving Ecuador's energy independence by 2030.**

***Index Terms*—Reconductoring, Generation, Planning, MILP, Ecuador.**

NOMENCLATURE

Sets

AL	Set of all branches i, j
CL	Number of candidate branches cl that can be added to the system, $CL \subset AL$
E	Set of existing branches el, $E \subset AL$
EG	Set of existing generators eg, $EG \subset G$
G	Set of all generators
N	Set of nodes i
NG	Set of candidate generators ng that can be added to the system, $NG \subset G$
R	Set of branches i, j that are candidates for reconductoring, $R \subset AL$
Y	Set of stages y

Parameters

α_y	Discount rate factor in stage y, (%)
δ_i	Phase angle of node i, (radians)
$\overline{\delta}_i$	Maximum phase angle of node i, (radians)
$\overline{n}_{i,j}$	Maximum number of branches that can be added to i, j
$\overline{Pl}_{i,j}$	Maximum power flow of branch i, j, (MW)
$Af_{y,g}$	Availability factor of generator g in stage y, (%)
c^{ens}	Cost of energy not served, ($/MWh)
$C^{ng}_{y,g}$	Investment cost of new generator g in stage y, (M$)
$C^{opg}_{y,g}$	Operating cost of generator g, in stage y ($/MWh)
$c^R_{i,j}$	Cost of reconductoring branch i, j, (MUSD)
$c^R_{i,j}n$	Cost of constructing a new branch identical to i, j post-reconductoring, (MUSD)
$cl_{i,j}$	Construction cost of a new branch i, j, (MUSD)
$n^E_{i,j}$	Number of existing branches i, j
$P^d_{y,i}$	Load of node i in stage y, (MW)
$P^{max}_{y,g}$	Maximum active power of generator g in stage y, (MW)
$P^{min}_{y,g}$	Minimum active power of generator g in stage y, (MW)
$Pl^R_{i,j}$	Maximum power flow of branch i, j after reconductoring, (MW)
Rm	Reserve margin, (MW)
Sb	Base power, (MVA)
Tb_i	Type of node i, (slack, PV, or PQ)
$x_{i,j}$	Reactance of branch i, j, (p.u.)
$x^R_{i,j}$	Reactance of branch i, j after reconductoring, (p.u.)

Variables

$\delta_{y,i}$	Phase angle of node i in stage y, (radians)
$P^{gen}_{y,g}$	Active power generated by generator g, in stage y, (MW)
$P_{y,i,j}$	Active power flow on new branch i, j, in stage y, (MW)
$P^E_{y,i,j}$	Active power flow on existing branch i, j, in stage y, (MW)
$PNS_{y,i}$	Power not served at node i, in stage y, (MW)
$Pr^E_{y,i,j}$	Active power flow on existing branch i, j, selected for reconductoring, in stage y, (MW)

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$Pr^{R}_{y,i,j}$ Power flow on new branch i,j after reconductoring the original branch, in stage y, (MW)

Binary Variables

$rpw_{y,i,j}$ Binary variable indicating if reconductoring is performed on existing branch i,j

$w^{gen}_{y,ng}$ Binary variable indicating if generator ng is constructed in stage y

$w^{line}_{y,i,j,cl}$ Binary variable indicating if a new transmission line is added to branch i,j in stage y

$w^{Rnew}_{y,i,j,cl}$ Binary variable indicating if a new line with the post-reconductoring characteristics is constructed for branch i,j in stage y

$z^{build}_{y,ng}$ Binary variable indicating if construction activities are initiated for generator ng in stage y

I. INTRODUCTION

The planning of electric system expansion faces increasingly significant challenges due to the growing energy demands in emerging and developing economies and the rapid adoption of advanced technologies with high electrical consumption in developed countries, including the expansion of electric vehicles and the integration of smart energy grids across various industrial sectors. According to the "Energy Outlook 2023", final electricity demand is projected to increase by approximately 75% by 2050 [1].

The efficiency and reliability of electrical systems are crucial for managing this increase in energy consumption. Annual costs associated with investment and operation represent only a fraction of the total costs of energy and auxiliary services traded each year. Given this, any failure in the generation or transmission system can have significant consequences for the stability of the electrical system, making it essential to have infrastructure that is not only efficient but also robust and strategically planned.

To address these challenges, planning entities utilize optimization models focused on minimizing costs, including investment, operation, and maintenance. The Generation Expansion Planning (GEP) problem addresses the identification of an investment schedule for new generation units, specifying their location and technology to meet the projected demand throughout the planning period [2]. On the other hand, Transmission Network Expansion Planning (TNEP) determines when, where, and how many new infrastructures, such as transmission lines, are necessary to meet reliability standards, economic efficiency, and sustainability [3].

Traditionally, GEP and TNEP have been treated as separate optimization problems, tackled by different market agents: producers in the case of GEP and regulated planners in the case of TNEP [4]. However, integrated planning for expanding generation and transmission (GTEP) provides significant added value, as transmission constraints can considerably influence the location and size of new generation units [5].

Research in TNEP has significantly evolved since the initial mixed integer linear disjunctive formulation proposed by Bahiense et al. [6]. The relaxation of the alternating current load flow equations presented in [7], allowed TNEP to address challenges related to reactive compensation [8], [9], security constraints [10], [11], resilience [12], distributed generation [13], and even GTEP models [14], [15]. The works of [16] and [17] tackle the TNEP problem from a different perspective, considering the repowering or reconductoring of existing transmission lines as a cost-effective alternative, given that under certain circumstances, it may be less costly to repower an existing transmission line rather than build a new circuit. For example, replacing ACSR conductors with HTLS can result in exceptional performance under a Dynamic Thermal Line Rating (DTLR) approach [18].

Although reconductoring has been discussed in the literature, to the best of the authors' knowledge, no prior studies have explored this approach for the Ecuadorian electricity system. This work develops a multistage joint generation and transmission planning model that uniquely considers the specific characteristics of Ecuador's energy mix and transmission network. The applicability of the model is demonstrated through the case study of the Ecuadorian power system, proposed as a test system (21-Bus-EC System), showcasing its potential impact and relevance to the country's electrical infrastructure.

The rest of this paper is structured as follows: Section 2 describes the mathematical model for generation and transmission expansion. It also outlines the data for constructing the proposed 21-Bus-EC System case study. Section 3 discusses the results obtained regarding the generation and transmission expansion strategy, along with total costs. Finally, Section 4 presents the conclusions and future research directions.

II. METHODOLOGY

This section describes the MILP model developed to co-optimize generation and transmission in the power grid, focusing specifically on reconductoring existing lines.

A. Formulation of the Problem

The problem involves optimizing the transmission network's efficiency by strategically reconductoring existing lines and integrating new generation resources where necessary. The MILP model considers the operational, construction, and upgrading costs associated with transmission and generation assets over a specified planning horizon.

1) Objective Function: The objective function, showing in Eq. (1), is designed to minimize the total cost of expanding and operating the transmission and generation network over a planning horizon. It incorporates the total discounted costs associated with the construction of new transmission lines $cl_{i,j}$, the reconductoring of existing lines $c^{R}_{i,j}$, and the operation of generators $c^{opg}_{y,g}$, as a function of their availability and energy demand throughout the year. Additionally, the costs derived from the investment in new generators c^{ng} and the cost of unsupplied energy c^{ens} are considered, reflecting the economic impact of not meeting the energy demand. The costs are adjusted by a discount factor, α_y, applied to each stage, reflecting the time value of money.

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$$\begin{aligned}\textbf{Minimize Cost} = \sum_{y}(\alpha_y \\ \cdot(\sum_{i,j,AL_{i,j}}(cl_{i,j}\cdot\sum_{cl}{w^{line}}_{y,i,j,cl}) \\ +\sum_{i,j,AL_{i,j}}({n^E}_{i,j}\cdot{c^R}_{i,j}\cdot rpw_{y,i,j}) \\ +\sum_{i,j,AL_{i,j}}({c^{Rnew}}_{i,j}\cdot\sum_{cl}{w^{Rnew}}_{y,i,j,cl}) \\ +\sum_{g}({c^{opg}}_{y,g}\cdot{p^{gen}}_{y,g}\cdot Af_{y,g}\cdot 8760*Sb) \\ +\sum_{g}({c^{ng}}_{y,g}\cdot{z^{build}}_{y,g}) \\ +\sum_{i}(PNS_{y,i}\cdot c^{ens}\cdot 8760*Sb)))\end{aligned} \tag{1}$$

2) Constraints: The MILP model is framed by a comprehensive set of constraints to ensure the system's operational feasibility and reliability across various scenarios. The constraints encapsulate load balancing (Eq. 2), which ensures that power generation matches demand at all nodes. Power flow constraints (Eqs. 3–6) regulate the electricity flow through existing and reconductored lines, ensuring that the transmission does not exceed physical and safety thresholds.

$$\begin{aligned}\sum_{i,AL_{i,j}}({P^E}_{y,i,j}+{Pr^E}_{y,i,j}+\sum_{cl}(P_{y,i,j,cl}+{Pr^R}_{y,i,j,cl})) \\ -\sum_{j,AL_{j,i}}({P^E}_{y,j,i}+{Pr^E}_{y,j,i} \\ +\sum_{cl}(P_{y,j,i,cl}+{Pr^R}_{y,j,i,cl}))+\sum_{g}({p^{gen}}_{y,g}) \\ +PNS_{y,i}=\frac{(1+Rm)\cdot \mathrm{Pd}_{y,i}}{Sb}; \qquad \forall y,i\end{aligned} \tag{2}$$

$$\begin{aligned}-(2\cdot\bar{\delta}_i\cdot rpw_{y,i,j}) \le \frac{{P^E}_{y,i,j}\cdot x_{i,j}}{{n^E}_{i,j}*Sb}-(\delta_{y,i}-\delta_{y,j} \\ \le 2\cdot\bar{\delta}_i\cdot rpw_{y,i,j};);\ \forall y,i,j \mid ({n^E}_{i,j}>0)\end{aligned} \tag{3}$$

$$\begin{aligned}-(2\cdot\bar{\delta}_i\cdot(1-rpw_{y,i,j})) \le \frac{{Pr^E}_{y,i,j}\cdot{x^R}_{i,j}}{{n^E}_{i,j}*Sb}-(\delta_{y,i}-\delta_{y,j} \\ \le 2\cdot\bar{\delta}_i\cdot(1-rpw_{y,i,j}); \\ \forall y,i,j \mid (({n^E}_{i,j}>0)\wedge({x^R}_{i,j}=0))\end{aligned} \tag{4}$$

$$\begin{aligned}-(2\cdot\bar{\delta}_i\cdot(1-{w^{line}}_{y,i,j,cl})) \le \frac{P_{y,i,j,cl}\cdot x_{i,j}}{Sb}-(\delta_{y,i}-\delta_{y,j} \\ \le 2\cdot\bar{\delta}_i\cdot(1-{w^{line}}_{y,i,j,cl}); \\ \forall y,i,j,cl\end{aligned} \tag{5}$$

$$\begin{aligned}-(2\cdot\bar{\delta}_i\cdot(1-{w^{Rnew}}_{y,i,j,cl})) \le \frac{{Pr^R}_{y,i,j,cl}\cdot{x^R}_{i,j}}{Sb} \\ -(\delta_{y,i}-\delta_{y,j}) \\ \le 2\cdot\bar{\delta}_i\cdot(1-{w^{Rnew}}_{y,i,j,cl}); \\ \forall y,i,j,cl \mid ({x^R}_{i,j}=0)\end{aligned} \tag{6}$$

Constraints on new line installations (Eqs. 7–10) control the introduction and operation of new infrastructure based on strategic needs and capacity limits. Generation capacity and operational safety are governed by Eqs. 18–19, imposing limits on the output of power plants to align with demand and safety standards. Further constraints (Eqs. 11–17, 20–23) address system-wide parameters such as voltage levels, phase angles, and the stability of power flows under varying conditions, ensuring comprehensive coverage of all technical aspects necessary for robust power system operation.

$$\begin{aligned}-({n^E}_{i,j}\cdot\frac{\overline{Pl}_{i,j}}{Sb}\cdot(1-rpw_{y,i,j})) \\ \le {P^E}_{y,i,j} \\ \le {n^E}_{i,j}\cdot\frac{\overline{Pl}_{i,j}}{Sb}\cdot(1-rpw_{y,i,j}); \qquad \forall y,i,j\end{aligned} \tag{7}$$

$$\begin{aligned}-({n^E}_{i,j}\cdot\frac{\overline{Pl}^R{}_{i,j}}{Sb}\cdot rpw_{y,i,j}) \\ \le {Pr^E}_{y,i,j} \\ \le {n^E}_{i,j}\cdot\frac{\overline{Pl}^R{}_{i,j}}{Sb}\cdot rpw_{y,i,j}; \qquad \forall y,i,j\end{aligned} \tag{8}$$

$$\begin{aligned}-({w^{line}}_{y,i,j,cl}\cdot\frac{\overline{Pl}_{i,j}}{Sb}) \le P_{y,i,j,cl} \\ \le {w^{line}}_{y,i,j,cl}\cdot\frac{\overline{Pl}_{i,j}}{Sb};\ \forall y,i,j,cl\end{aligned} \tag{9}$$

$$\begin{aligned}-({w^{Rnew}}_{y,i,j,cl}\cdot\frac{\overline{Pl}^R{}_{i,j}}{Sb}) \le {Pr^R}_{y,i,j,cl} \\ \le {w^{Rnew}}_{y,i,j,cl}\cdot\frac{\overline{Pl}^R{}_{i,j}}{Sb};\ \forall y,i,j,cl\end{aligned} \tag{10}$$

$$\sum_{y,cl}{w^{line}}_{y,i,j,cl}\le\overline{n}_{i,j}; \qquad \forall i,j \tag{11}$$

$${w^{line}}_{y,i,j,cl}\le({w^{line}}_{y,i,j,cl-1});\ \forall y,i,j,cl \mid (\mathrm{cl})>1) \tag{12}$$

$$\sum_{y,cl}{w^{Rnew}}_{y,i,j,cl}\le\overline{n}_{i,j}; \qquad \forall i,j \tag{13}$$

$${w^{Rnew}}_{y,i,j,cl}\le{w^{Rnew}}_{y,i,j,cl-1};\ \forall y,i,j,cl \mid (\mathrm{cl})>1) \tag{14}$$

$${w^{Rnew}}_{y,i,j,cl}\le rpw_{y,i,j}; \qquad \forall y,i,j,cl \tag{15}$$

$${w^{line}}_{y,i,j,cl}\le 1-rpw_{y,i,j}; \qquad \forall y,i,j,cl \tag{16}$$

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$$rpw_{y,i,j} = 0; \qquad \forall y, i, j, cl \mid ({x^R}_{i,j} = 0) \quad (17)$$

$$\frac{{P^{min}}_{y,eg}}{Sb} \cdot Af_{y,g} \leq {p^{gen}}_{y,eg} \leq \frac{{P^{max}}_{y,eg}}{Sb} \cdot Af_{y,g}; \quad \forall y, eg \quad (18)$$

$$\frac{{P^{min}}_{y,ng}}{Sb} \cdot {w^{gen}}_{y,ng} \cdot Af_{y,g} \leq {p^{gen}}_{y,ng} \leq \frac{{P^{max}}_{y,ng}}{Sb} \cdot {w^{gen}}_{y,ng} \cdot Af_{y,g}; \qquad \forall y, ng \quad (19)$$

$$-\bar{\bar{\delta}}_i \leq \delta_{y,i} \leq \bar{\bar{\delta}}_i; \qquad \forall y, i \quad (20)$$

$$\sum_y w^{gen}_{y,ng} \leq 1 \quad ; \forall ng \quad (21)$$

$$w^{gen}_{y,ng} - w^{gen}_{y-1,ng} \geq 0 \quad ; \forall y > 1, ng \quad (22)$$

$$w^{gen}_{y,ng} - w^{gen}_{y-1,ng} = z^{build}_{y,ng} \quad ; \forall y > 1, ng \quad (23)$$

B. Case Study: 21-Bus-EC System

Ecuador's electricity system relies heavily on hydroelectricity. In 2023, of the total effective power of 8,254.45 MW, hydroelectric plants constituted 62.42%. That year, the total net energy production was 31,903.26 GWh, composed of 59.02% hydroelectricity, 26.2% thermal sources, 1.7% non-conventional renewable sources, and 8.1% imported electricity from Colombia and Peru. Additionally, 4.7% of the energy demand was not satisfied due to load-shedding protocols implemented by the system operator [19].

This strong dependence on hydroelectricity makes the system particularly vulnerable to water resource variability, exacerbated by drought scenarios linked to climate change. This vulnerability was exposed in 2023, when electricity imports from Colombia reached $285.88 MUSD, a figure ten times higher than in 2022, highlighting the critical need to diversify energy sources and strengthen the system's resilience to severe droughts [20].

The MILP model proposed in Section A will be used to investigate and address the above-mentioned challenges. The applicability of the model is illustrated by a simplified 21-node network representing the National Interconnected System of Ecuador, referred to as the 21-Bus-EC System. This network includes a 230 kV transmission ring with 15 nodes, two 138 kV nodes, and a 500 kV backbone system with four additional nodes, as shown in Figure 1. To ensure the accuracy of power flow calculations within this network, the results were validated using DigSilent PowerFactory, confirming the reliability and effectiveness of the proposed modeling approach.

The planning scenario for expanding Ecuador's electricity system extends from 2024 to 2030 maintaining a reserve margin of 15%. A multivariate linear regression model was used to determine the maximum demand to be satisfied, using GDP and population projections showing in Table I as predictor variables. The projections reveal that the demand for electric power will increase by 14.53% by the year 2030. Then, this demand is distributed among the 21 nodes according to a participation factor designated for each node as a parameter.

Table I
PROJECTED PEAK DEMAND FOR ECUADOR

Year	GDP [MUSD]	Population [thousands]	Demand [MW]
2024	114,486.07	18,508.50	4,546.52
2025	116,612.36	18,757.68	4,656.63
2026	118,738.66	19,006.86	4,766.74
2027	120,864.95	19,256.04	4,876.86
2028	122,991.25	19,505.22	4,986.97
2029	125,117.54	19,754.40	5,097.08
2030	127,243.84	20,003.58	5,207.19

*GDP values are expressed in constant 2010 USD.

The data used to model the thechnical and economic characteristics of the generators and loads were obtained from open-access sources, including the Electricity Master Plan (PME) [21], reports from the electricity system operator CENACE [19], ARCERNRR [20] report and Lazard's Levelized Cost of Energy Analysis report [22].

The subset of candidate generation facilities includes three hydroelectric plants as proposed in the PME: Santiago, Cardenillo, and Toachi Pilatón. Furthermore, four generic generation plants are outlined in Table II , which includes two thermal and two photovoltaic facilities strategically positioned at the Totoras, Pascuales and Santa Rosa nodes to meet the high energy demands of Ecuador's largest cities, Ambato, Guayaquil and Quito, respectively. These generation facilities are assumed to be integral components of the strategies devised by planners to mitigate the energy crisis in the short term.

To quantitatively assess the impact of climatic variations on hydroelectric generation, this study incorporates an availability factor that reflects the variability in water resources. This factor adjusts the expected output from hydroelectric facilities. It is applied to compensate for both seasonal fluctuations and long-term climatic trends, ensuring that the model accurately reflects real-world conditions. This approach is thoroughly documented in the recent academic literature, including references [23] and [24].

On the other hand, the reconductoring approach involves the replacement of existing transmission line conductors (ACSR Bluejay 1113 conductor) with higher capacity conductors (ACSS Bluejay 1113). The latter offers an ampacity 22% higher than the original conductor, as cited in [17]. This change is intended to increase the efficiency and capacity of these specific lines. Table III illustrates the lines selected for reconductoring.

The optimization model was developed using the General Algebraic Modeling System (GAMS) and solved with the

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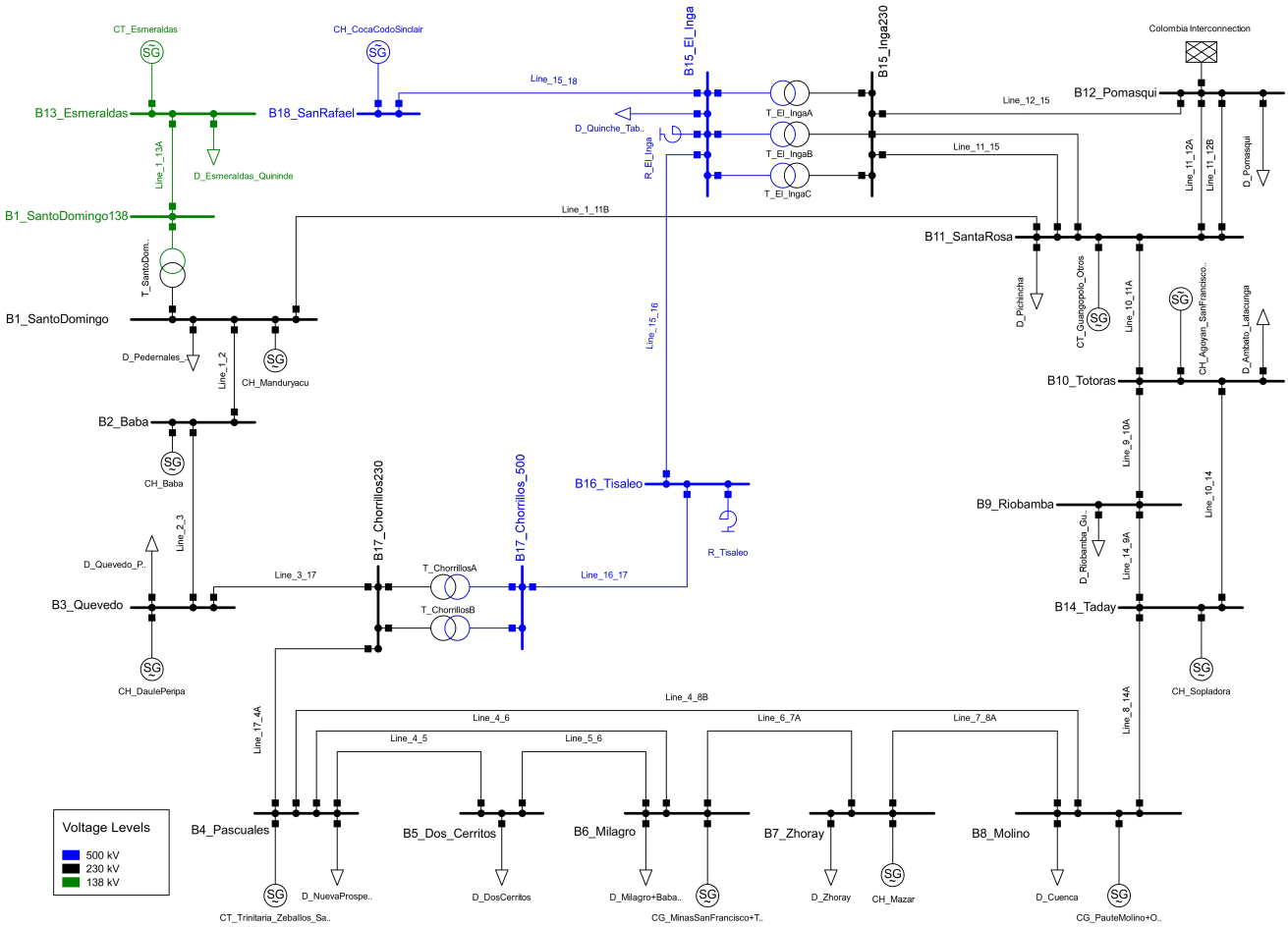


Figure 1. Representation of the National Interconnected System of Ecuador as a 21-bus network.

Table II
CANDIDATE POWER PLANTS PARAMETERS

Candidate Power Plant	$P^{max}_{y,g}$ **[MW]**	$C^{opg}_{y,g}$ **[MUSD]**	$C^{ng}_{y,g}$ **[MUSD]**	$Af_{y,g}$ **[%]**
N_HP_ToachiPilaton	254.00	2.00	100.00	60.00
N_TP_Generic_Gas	300.00	47.50	350.00	85.00
N_HP_Cardenillo	596.00	2.00	1300.00	60.00
N_Photovoltaic_Generic1	200.00	0.00	300.00	25.00
N_TP_Generic_FuelOil	300.00	37.93	350.00	80.00
N_HP_Santiago	2400.00	2.00	3000.00	60.00
N_Photovoltaic_Generic2	200.00	0.00	300.00	25.00

Table III
CANDIDATE LINES FOR RECONDUCTORING

from node	**to node**	$n^E_{i,j}$	$\mathbf{PI}^R_{i,j}$ **[MW]**
B14_Taday	B10_Totoras	1	322.99
B12_Pomasqui	B11_SantaRosa	2	303.78
B9_Riobamba	B14_Taday	2	312.93
B1_SantoDomingo	B11_SantaRosa	2	312.93
B2_Baba	B3_Quevedo	1	322.99
B4_Pascuales	B6_Milagro	1	322.99
B5_Dos_Cerritos	B6_Milagro	1	322.99
B7_Zhoray	B8_Molino	1	312.75

Gurobi solver. All computational tasks were executed on a high-performance computing platform equipped with an Intel i9-13900K processor and 128 GB of RAM. To quantify the computational demands of the model, the average CPU time recorded for solving the base scenario was approximately 11.52 seconds. Table IV provides a detailed summary of the model's computational statistics.

Table IV
SUMMARY OF OPTIMIZATION MODEL STATISTICS

Parameter	**Value**	**Parameter**	**Value**
Blocks of equations	35	Single equations	110.116
Blocks of variables	13	Single variables	41.294
Non zero elements	216.262	Discrete variables	18.613

III. RESULTS

The application of the proposed optimization model to the 21-Bus-EC System case study reveals that the budget required to address Ecuador's energy crisis through 2030 is projected to be $7169.31 MUSD. It is crucial to note that the variable costs applied in the model provide an approximation and do

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not necessarily reflect the exact operational reality of each power plant. Consequently, the model results are intended to provide decision support and not to define the planner's policy actions directly.

The cost breakdown for the period 2024-2030 is detailed in Table V. Investments in six new power plants account for 59.33% of the total budget. Costs associated with the operation of existing power plants represent 38.38%, underscoring the continuous need for maintaining and optimizing current assets. A smaller portion of the budget is allocated to the transmission system, including the construction of new lines and the reconductoring of existing lines, which collectively account for approximately 2.07% of the total budget.

Table V
COST BREAKDOWN FOR THE PERIOD 2024-2030

Cost Item	Cost [MUSD]
Power Not Served	0.00
New Line Built	125.98
Reconductoring	35.58
New Line post-reconductoring	2.07
New Power Plants	4253.77
Operational Power Plants	2751.92

As shown in Figure 2, the 2024-2030 period underlines a sustained dependence on hydropower. By 2030, hydropower is expected to cover 72.01% of Ecuador's electricity demand. Thermal power plants are expected to contribute 25.26%, while photovoltaic generation will contribute a modest 1.75%. It is worth noting that imports from Colombia drop from 8.69% to 0.99% at the end of the planning period, indicating that even with investments in electricity generation, dependence on generation from Colombia would remain, especially in Ecuador's dry season, where availability factors tend to be lower.

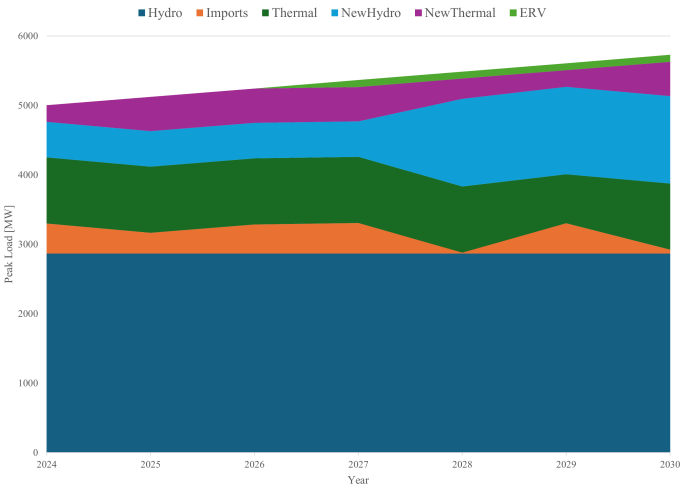


Figure 2. Ecuador's electricity mix for the period 2024-2030.

Although reconductoring represents a minimal cost in the total investment, the results presented in Fig. 3 suggest that increasing the transmission capacity of existing lines may result in a more cost-efficient solution.

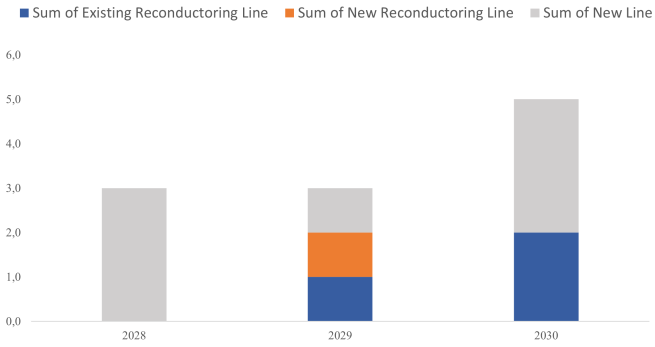


Figure 3. New Transmission Lines for the period 2024-2030.

CONCLUSIONS

This study presents optimal planning for the expansion of Ecuador's electrical system using a MILP model that incorporates a reconductoring approach. Key findings from the application of this model include:

The MILP model effectively integrates reconductoring strategies within a multistage planning framework for generation and transmission systems. By accounting for the comprehensive costs of construction, reconductoring, and operational dynamics, the model strategically minimizes the total cost of electricity supply over multiple planning periods.

The application of the MILP model to the 21-Bus-EC System demonstrates its utility in planning under Ecuador's heavy reliance on hydropower. The model indicates necessary investments totaling $4,253.77 MUSD for new power plants by 2030, emphasizing the shift towards diversifying energy sources. Additionally, $163.33 MUSD is allocated for enhancing the transmission system, including new constructions and reconductoring. This represents a strategic effort to reduce the system's dependence on imported electricity, which is expected to decrease to 0.99% of the energy mix by 2030.

Future research will aim to enhance the model by incorporating the stochastic nature of water resources, and climate change factors. These enhancements will allow the model to provide more comprehensive solutions that align with sustainable development goals.

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